



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SRI VINAYAGA AGENCY

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

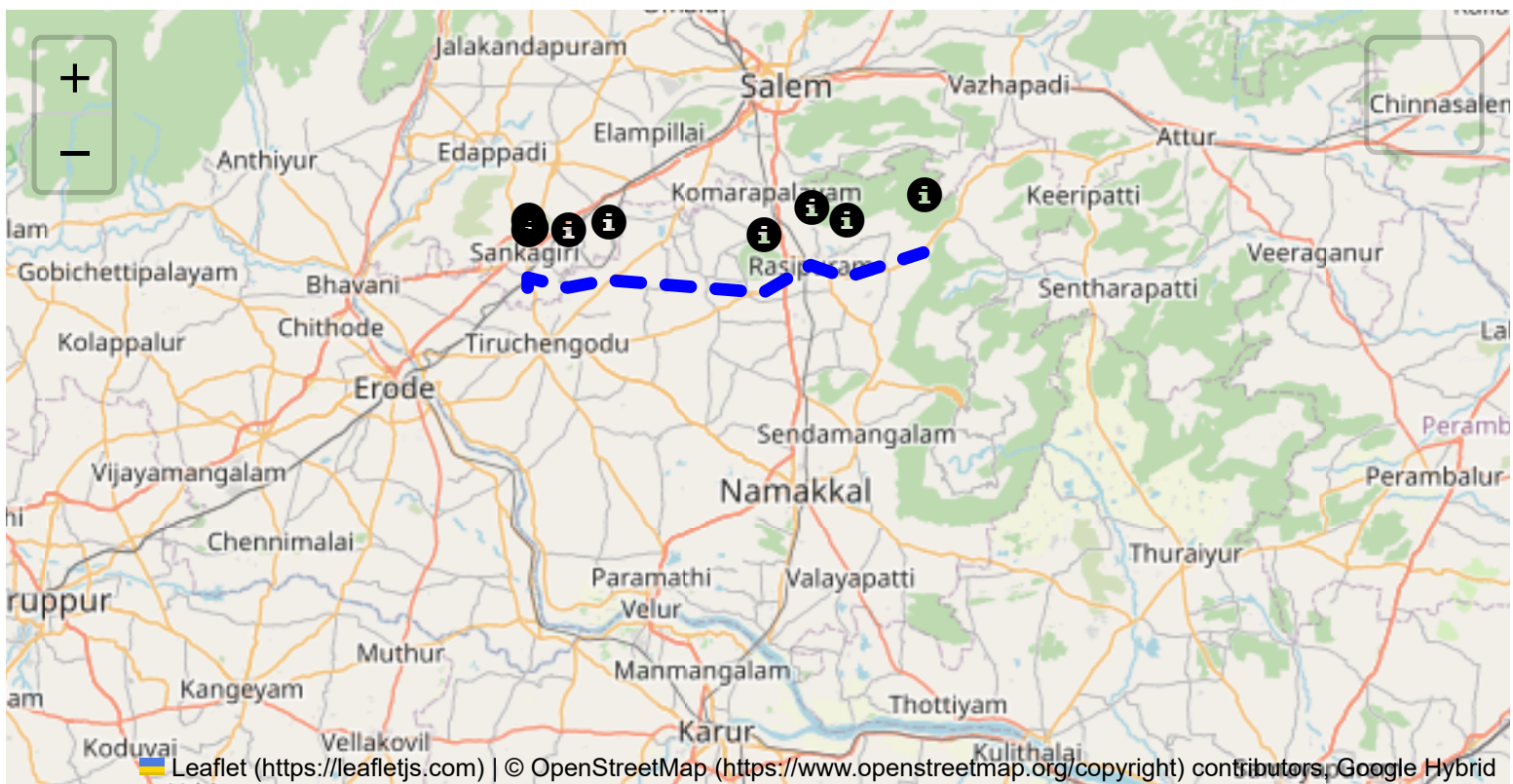
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

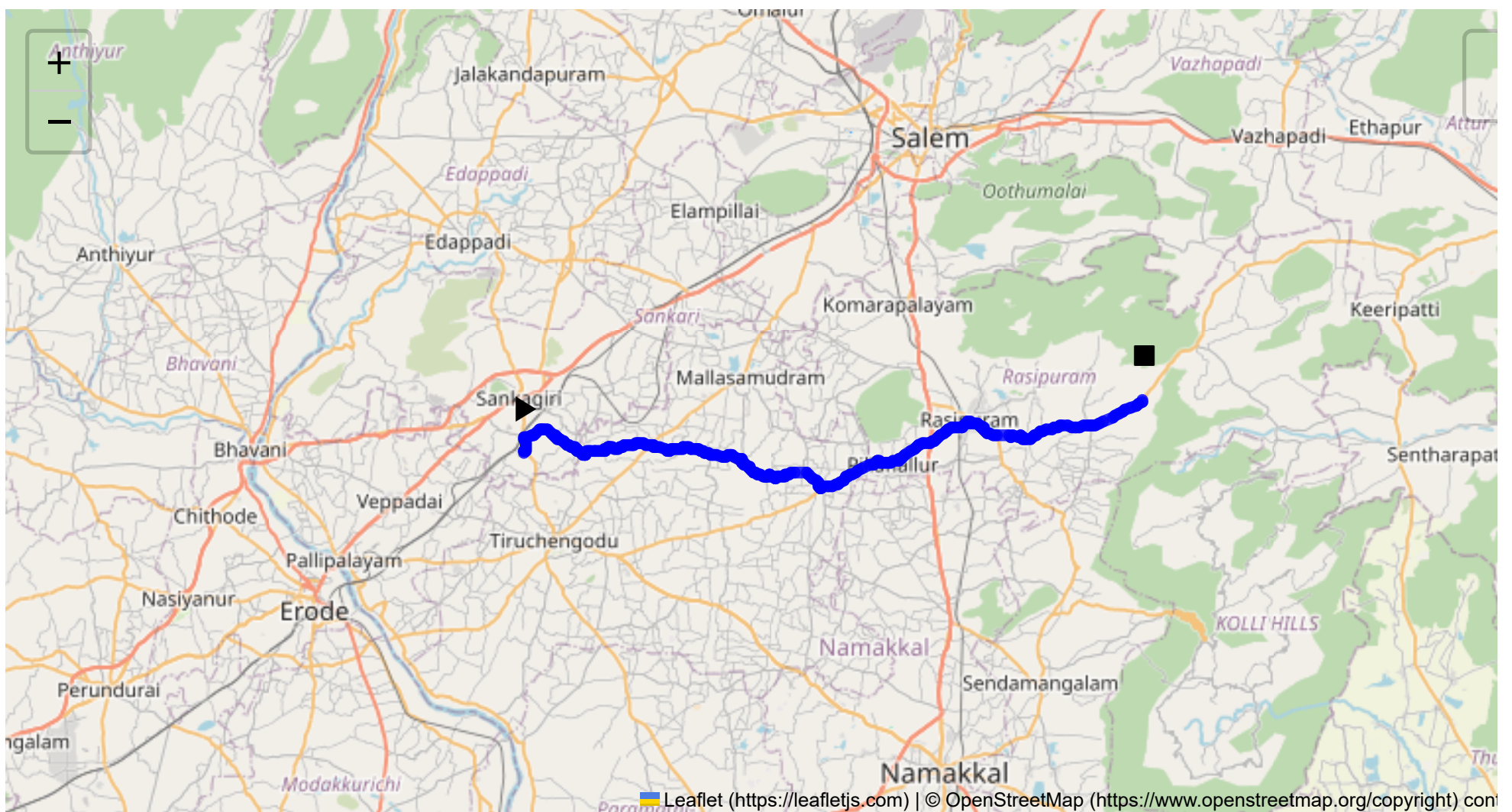
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 55.44 km
Estimated Duration: 1.5 hours
Adjusted Duration (Heavy Vehicle): 1.9 hours
Start: (11.4381, 77.8734)
End: (11.474535, 78.306641)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map: The route from Sangagiri to Rasipuram in Tamil Nadu spans approximately 55.44 kilometers. It travels through a mix of rural and semi-urban areas, taking the driver through notable waypoints such as Kangeyampalayam, Kozhikkalnatham, and Pillanatham. The primary roads involved are local roads with connections to the Tiruchengode - Salem Main Road and SH 79. This route navigates through agricultural regions and small-town infrastructures.

2. Typical Weather Conditions and Potential Hazards: The region experiences a tropical climate with hot summers, moderate winters, and monsoon rains. Summers (March to May) can average around 35°C, potentially affecting vehicle performance. Monsoon season (June to September) can lead to water-logging and reduced road traction, increasing the likelihood of accidents. Potential hazards include slippery roads during rain and reduced visibility due to fog in early mornings or late evenings, especially during the winter months.

3. Analysis of Traffic Patterns: Traffic density tends to increase during the morning (7 AM - 10 AM) and evening hours (5 PM - 8 PM), particularly in areas near Tiruchengode and Rasipuram due to local business activities and school commutes. Farmers' markets and local bazaars could also contribute to congestion, especially during weekends or market days.

4. Assessment of Road Quality and Infrastructure: The roads vary from well-paved highways to narrower country roads. The Tiruchengode - Salem Main Road (near Pillanatham) is generally in good condition with adequate signage, but local roads may have occasional potholes, sharp turns, and limited shoulder space. Some rural sections may lack proper lighting at night.

5. Suggestions for Alternative Routes in Emergencies: In cases of route blockage or emergency, alternative routes can include using NH 44 or the Salem-Namakkal Road (NH 7) to bypass local congestion. These highways generally offer better road conditions and faster travel due to higher speed limits and less impacted by local traffic.

6. Summary of Local Regulations Affecting Hazardous Material Transport: Transport of hazardous materials is subject to strict regulations. Nighttime movement is often restricted. Checkpoints at towns can enforce compliance with safety certificates, and transport permits must be maintained. Adherence to safety protocols such as having proper placarding, spill kits, and driver training is mandatory.

7. Overview of Historical Incidents: Data on historical incidents can be sparse, but any available records suggest occasional issues involve brake failures on slopes and incidents during the rainy season. Detailed incident databases from local transport authorities would be needed for a thorough historical analysis.

8. Environmental Considerations and Sensitive Areas: The route traverses several agricultural areas and small ecosystems. Care must be taken when traveling near water bodies and plantations due to runoff concerns in case of spills. Awareness of farming activities and livestock is essential in rural areas.

9. Analysis of Communication Coverage: The route generally maintains good mobile coverage with some potential dead zones in rural stretches between Kozhikkalnatham and Pillanatham. Critical communication equipment and offline maps would be prudent in cases of coverage loss.

10. Estimated Emergency Response Times: Given the proximity to roads and towns, emergency response times in urban sections can range from 15-30 minutes. However, in rural stretches, it might extend to 45 minutes or more, especially in low-visibility conditions or if access is hampered by traffic or road quality.

12. Overall Summary of Risk Assessment: The route poses several risks typical of semi-urban and rural transport routes in India. Weather-induced road conditions, traffic congestion in towns, and potential infrastructure limitations are primary concerns. Preparedness for variable conditions, strict adherence to transport regulations, and efficient communication and emergency planning are critical for ensuring safe passage for hazardous materials. Regular updates on road conditions and weather forecasts should be monitored to mitigate risks effectively.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44028, 77.87541	15 KM/Hr
2	Turn	High	11.44852, 77.87432	15 KM/Hr
3	Turn	Medium	11.45470, 77.88943	30 KM/Hr
4	Turn	Medium	11.44194, 77.90923	30 KM/Hr
5	Turn	High	11.43673, 77.91697	15 KM/Hr
6	Turn	Medium	11.43690, 77.91706	30 KM/Hr
7	Turn	High	11.43928, 77.91712	15 KM/Hr
8	Turn	Medium	11.43969, 77.92357	30 KM/Hr
9	Turn	Medium	11.43996, 77.92574	30 KM/Hr
10	Turn	Medium	11.43982, 77.93018	30 KM/Hr
11	Turn	Medium	11.44033, 77.93076	30 KM/Hr
12	Turn	High	11.44071, 77.93211	15 KM/Hr
13	Turn	Medium	11.44135, 77.93220	30 KM/Hr
14	Turn	Medium	11.44175, 77.93249	30 KM/Hr
15	Turn	High	11.44248, 77.93247	15 KM/Hr
16	Turn	Medium	11.44289, 77.93420	30 KM/Hr
17	Turn	Medium	11.44094, 77.93824	30 KM/Hr
18	Turn	Medium	11.44547, 77.95309	30 KM/Hr
19	Turn	Medium	11.44475, 77.95777	30 KM/Hr
20	Turn	Medium	11.44370, 77.96020	30 KM/Hr
21	Turn	Medium	11.44395, 77.96298	30 KM/Hr
22	Turn	High	11.44307, 77.96476	15 KM/Hr
23	Turn	Medium	11.44248, 77.96497	30 KM/Hr
24	Turn	Medium	11.44231, 77.96525	30 KM/Hr
25	Turn	Medium	11.44155, 77.98447	30 KM/Hr
26	Turn	Medium	11.43999, 77.99860	30 KM/Hr
27	Turn	High	11.43524, 78.01527	15 KM/Hr
28	Turn	Medium	11.43718, 78.01764	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
29	Turn	Medium	11.43718, 78.01789	30 KM/Hr
30	Turn	High	11.43435, 78.02651	15 KM/Hr
31	Turn	Medium	11.43256, 78.02640	30 KM/Hr
32	Turn	Medium	11.42461, 78.03453	30 KM/Hr
33	Turn	High	11.42323, 78.03841	15 KM/Hr
34	Turn	Medium	11.42332, 78.03862	30 KM/Hr
35	Turn	High	11.42367, 78.03885	15 KM/Hr
36	Turn	High	11.42215, 78.04279	15 KM/Hr
37	Blind Spot	Blind Spot	11.42263, 78.04320	10 KM/Hr
38	Blind Spot	Blind Spot	11.42152, 78.04471	10 KM/Hr
39	Turn	High	11.42316, 78.04536	15 KM/Hr
40	Turn	Medium	11.42316, 78.04551	30 KM/Hr
41	Turn	Medium	11.42117, 78.05023	30 KM/Hr
42	Turn	High	11.42203, 78.05577	15 KM/Hr
43	Turn	High	11.42274, 78.05597	15 KM/Hr
44	Turn	Medium	11.42447, 78.06053	30 KM/Hr
45	Turn	High	11.42430, 78.07221	15 KM/Hr
46	Turn	Medium	11.41946, 78.07856	30 KM/Hr
47	Turn	Medium	11.41672, 78.07954	30 KM/Hr
48	Turn	High	11.41467, 78.08053	15 KM/Hr
49	Turn	High	11.41463, 78.08067	15 KM/Hr
50	Turn	Medium	11.41475, 78.08087	30 KM/Hr
51	Turn	High	11.43257, 78.12096	15 KM/Hr
52	Turn	Medium	11.44496, 78.15264	30 KM/Hr
53	Turn	High	11.45597, 78.18073	15 KM/Hr
54	Turn	Medium	11.45728, 78.18115	30 KM/Hr
55	Blind Spot	Blind Spot	11.46034, 78.18528	10 KM/Hr
56	Turn	High	11.45909, 78.18553	15 KM/Hr
57	Turn	Medium	11.45858, 78.18772	30 KM/Hr
58	Turn	Medium	11.45823, 78.18818	30 KM/Hr
59	Turn	Medium	11.44994, 78.22115	30 KM/Hr
60	Turn	High	11.44925, 78.22288	15 KM/Hr
61	Turn	Medium	11.44813, 78.22305	30 KM/Hr
62	Turn	Medium	11.44770, 78.22363	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
7	hospital	Sudha Hospital	11.4555773, 78.1760125	30 km/h	Medium
8	hospital	Government Hospital Blood Bank, Rasipuram	11.4612082, 78.1849249	30 km/h	Medium
9	hospital	M.R. Hospital, Namakkal	11.4613462, 78.1861702	30 km/h	Medium
10	police	Police Station Rasipuram	11.4617825, 78.1847411	30 km/h	Medium
11	hospital	Government Hospital, Rasipuram	11.4612987, 78.1851369	30 km/h	Medium
12	hospital	Rasi Hospital	11.4618368, 78.1855695	30 km/h	Medium
13	hospital	dr.thangavel dch hospital	11.4619955, 78.1849181	30 km/h	Medium
14	hospital	surya hospital	11.4622163, 78.1840544	30 km/h	Medium
15	hospital	R K Eye Care Center	11.4589079, 78.1913596	30 km/h	Medium
16	hospital	Manickkam Hospital	11.4568944, 78.1926061	30 km/h	Medium
17	hospital	Snathi Hospital	11.4564245, 78.1932485	30 km/h	Medium
18	hospital	Sugam Hospital,Rasipuram	11.4553007, 78.1936221	30 km/h	Medium
20	hospital	Government Hospital, Namagiripettai	11.4569484, 78.265263	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	Sri Vinayaga Matriculation Higher Secondary School	11.4233682, 78.0736517	30 km/h	Medium
1	marketplace	Farmer Market	11.4146849, 78.0805463	30 km/h	Medium
2	school	Government Higher Secondary School	11.4150045, 78.0838172	30 km/h	Medium
3	school	Vidya madir school	11.4356335, 78.1381408	30 km/h	Medium
4	college	Thiruvalluvar Govt arts college	11.4449352, 78.1595562	30 km/h	Medium
5	school	Venkateswara School	11.4468258, 78.1595761	30 km/h	Medium
6	school	Venkateswara Middle school	11.4472063, 78.1596191	30 km/h	Medium
19	school	Govt. Hr. Sec. School	11.4585324, 78.1933666	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44028, 77.87541



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44852, 77.87432



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45470, 77.88943



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44194, 77.90923



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.43673, 77.91697



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.43690, 77.91706



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.43928, 77.91712



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.43969, 77.92357



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.43996, 77.92574



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.43982, 77.93018



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44033, 77.93076



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44071, 77.93211



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44135, 77.93220



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44175, 77.93249



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44248, 77.93247



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44289, 77.93420



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44094, 77.93824



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44547, 77.95309



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44475, 77.95777



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44370, 77.96020



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44395, 77.96298



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44307, 77.96476



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44248, 77.96497



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44231, 77.96525



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44155, 77.98447



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.43999, 77.99860



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.43524, 78.01527



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.43718, 78.01764



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.43718, 78.01789



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.43435, 78.02651



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.43256, 78.02640



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.42461, 78.03453



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.42323, 78.03841



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.42332, 78.03862



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.42367, 78.03885



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.42215, 78.04279



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.42263, 78.04320



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.42152, 78.04471



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.42316, 78.04536



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.42316, 78.04551



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.42117, 78.05023



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.42203, 78.05577



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.42274, 78.05597



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.42447, 78.06053



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.42430, 78.07221



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41946, 78.07856



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41672, 78.07954



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.41467, 78.08053



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.41463, 78.08067



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41475, 78.08087



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.43257, 78.12096



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44496, 78.15264



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45597, 78.18073



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45728, 78.18115



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.46034, 78.18528



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45909, 78.18553



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45858, 78.18772



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45823, 78.18818



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44994, 78.22115



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44925, 78.22288



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44813, 78.22305



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44770, 78.22363
